Practice Set: Unit III

Continuous Distributions and Transformations

STA201 – Theory of Probability and Statistics

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Instructions

- Use the standard normal table.
- Round all probabilities to 4 decimal places unless stated otherwise.

Questions

1. Evaluate the following using the gamma function:

$$\text{(a) } \Gamma(6), \qquad \text{(b) } \Gamma\left(\frac{7}{2}\right), \qquad \text{(c) } \int_0^\infty x^5 e^{-2x}\, dx.$$

2. Buses on a route arrive exactly every 30 minutes. A passenger arrives at the bus stop at a completely random time, so the waiting time W (in minutes) is uniform on $[0, 30]$.

   (a) Find the expected waiting time and its variance.
   (b) Find the probability that the passenger waits more than 20 minutes.

3. The lifetime T (in hours) of an industrial fan is exponentially distributed with mean 2000 hours.

   (a) Find $P(T > 1500)$.
   (b) Find $P(1000 < T < 2000)$.

4. A component has an exponential lifetime with mean 1000 hours. Suppose the component has already operated for 500 hours without failure.

   (a) Find the probability that it survives at least another 1000 hours.
   (b) State the property of the exponential distribution that you used.

5. The lifetime of a device (in years) is exponential with rate $\lambda = 0.2$.

   (a) Find the median lifetime.
   (b) Compare it with the mean. Explain the reason for the difference.

6. A parallel system has two independent components with lifetimes $T_1 \sim \text{Exp}(0.03)$ and $T_2 \sim \text{Exp}(0.02)$. The system fails as soon as the first component fails.

   (a) Identify the distribution of the system lifetime and give its parameter.
   (b) Find the probability that the system lasts more than 20 hours.
   (c) Find the expected system lifetime.

7. Breakdowns of a machine follow a Poisson process at the rate of 2 per hour. Let X be the time (in hours) until the third breakdown.

   (a) Identify the distribution of X (shape and scale parameters).
   (b) Using the CDF formula for integer shape, compute $P(X \leq 4)$.

8. A random variable X has PDF

$$f(x) = 6x(1-x), \qquad 0 < x < 1.$$

   (a) Identify the distribution (state α and β).
   (b) Find $E[X]$ and $\mathrm{Var}(X)$.

9. Let $X \sim \mathrm{Beta}(2, 2)$ with PDF $f(x) = 6x(1-x)$, $0 < x < 1$. Find $P(X < 0.3)$.

10. Let $X \sim \mathrm{Beta}(2, 2)$ with PDF $f(x) = 6x(1-x)$, $0 < x < 1$.

    (a) Find the mode.
    (b) Find $P(0.4 < X < 0.6)$.

11. Let $X \sim N(60, 100)$ (variance 100). Find $P(X < 75)$.

12. Let $X \sim N(60, 100)$. Find $P(50 < X < 70)$.

13. A rope's breaking strength X is normally distributed with standard deviation 5. It is found that 90% of ropes have strength above 30.

    (a) Find the mean strength μ.

14. A packaging machine fills bags with a normally distributed weight having mean 200 g. It is found that 5% of bags contain less than 180 g.

    (a) Find the standard deviation σ.

15. Two independent resistors have resistances $R_1 \sim N(30, 9)$ and $R_2 \sim N(40, 16)$, where the second parameter is the variance.

    (a) Give the distribution of the total resistance $R_1 + R_2$.
    (b) Find $P(R_1 + R_2 > 80)$.

16. A binomial random variable $X \sim B(n = 100,\ p = 0.3)$ is approximated by a normal distribution.

    (a) State the mean and standard deviation of the approximating normal distribution.
    (b) Using the continuity correction, estimate $P(X \leq 35)$.

17. A random variable X is uniformly distributed on $\{-2, -1, 0, 1, 2\}$, so $p_X(x) = 1/5$ for each x. Define $Y = 4X + 3$.

    (a) Find the PMF of Y.
    (b) Compute $E[Y]$ and $\mathrm{Var}(Y)$.

18. Let $X \sim \mathrm{Poisson}(4)$. Define $Y = 3X$.

    (a) Find the PMF of Y.
    (b) Compute $P(Y = 6)$.
    (c) Compute $E[Y]$ and $\mathrm{Var}(Y)$.

19. Let $X \sim \text{Binomial}(8, 0.4)$. Define $Y = 8 - X$.

    (a) Find the PMF of Y.
    (b) State the distribution of Y.
    (c) Compute $E[Y]$ and $\text{Var}(Y)$.

20. Let $X \sim \text{Geometric}(p)$ with $p = 0.2$, so $p_X(x) = (0.8)^{x-1}(0.2)$ for $x = 1, 2, 3, \ldots$. Define $Y = X + 4$.

    (a) Find the PMF of Y.
    (b) Compute $P(Y \leq 6)$.
    (c) Compute $E[Y]$ and $\text{Var}(Y)$.